

(a) Twin construction A: dog B: rooster



Reverse query $\leftrightarrow$ **swap target**

q_+ : dog followed by rooster $\rightarrow W_+$

q_- : rooster followed by dog $\rightarrow W_-$

(b) RelTwin training

Stage 1: Answer learning

	$v(W_+)$	$v(W_-)$
q_+	$s_{++} \uparrow$	$s_{+-} \downarrow$
q_-	$s_{-+} \downarrow$	$s_{--} \uparrow$

Competition + reversal consistency

Stage 2: Set preferences



Coverage · count · boundaries

Inference Audio + query $\rightarrow$ Adapted backbone $\rightarrow$ Timestamps

(c) Window selection

